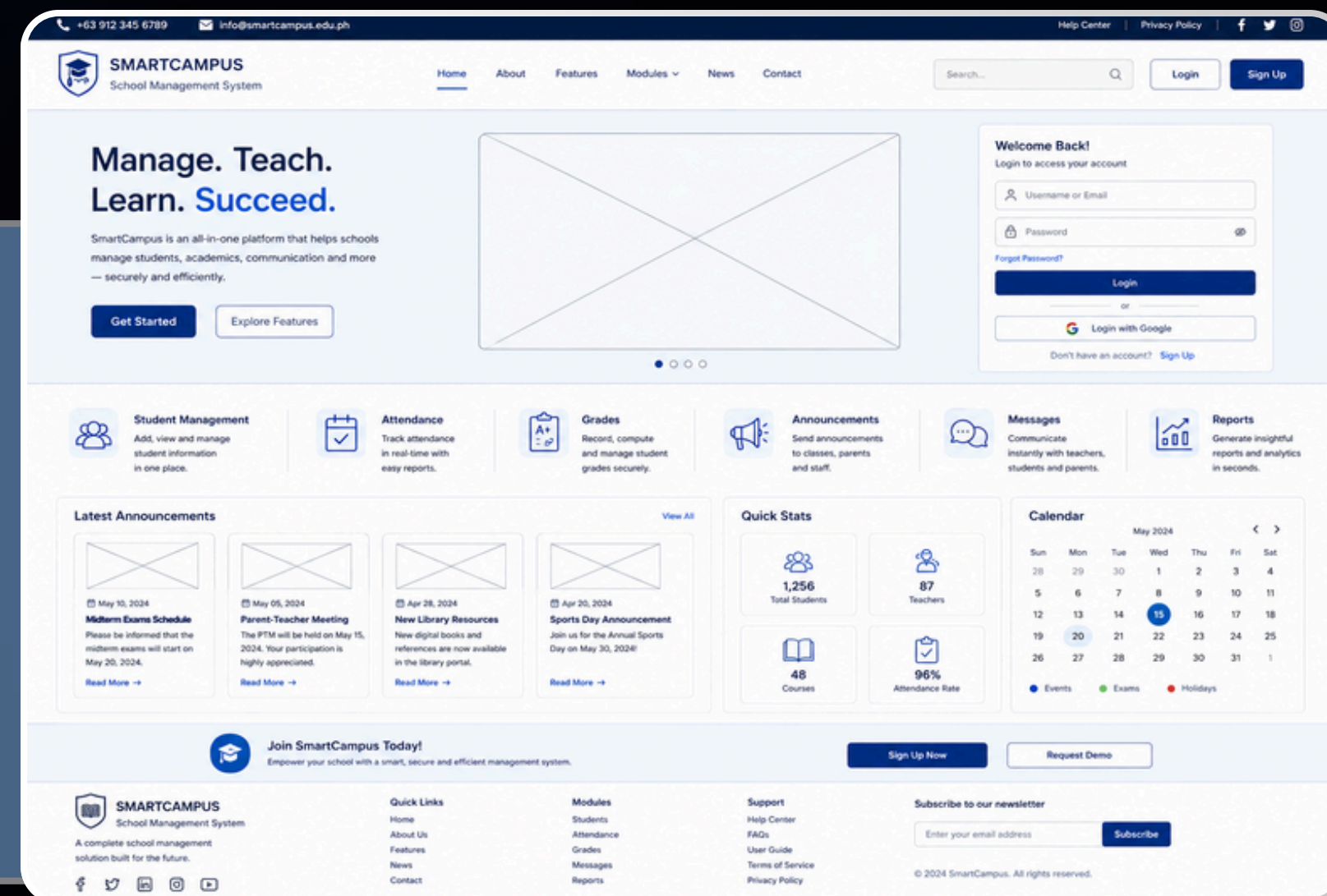


SmartCampus School Management System

System Overview

SmartCampus is a cloud-based School Management System that manages:

- Student enrollment
- Attendance monitoring
- Online grade submission
- Parent-teacher communication
- Digital school records



Main Users

- School Administrators
- Teachers
- Students
- Parents

Security Focus

Protecting academic records and preventing unauthorized grade manipulation

Bonifacio, Shane Leonard L.
Ferrer, Jan Lambert E.
Jay Mark Mortel
Pablo Jhanric Hanamichi

Security Requirements

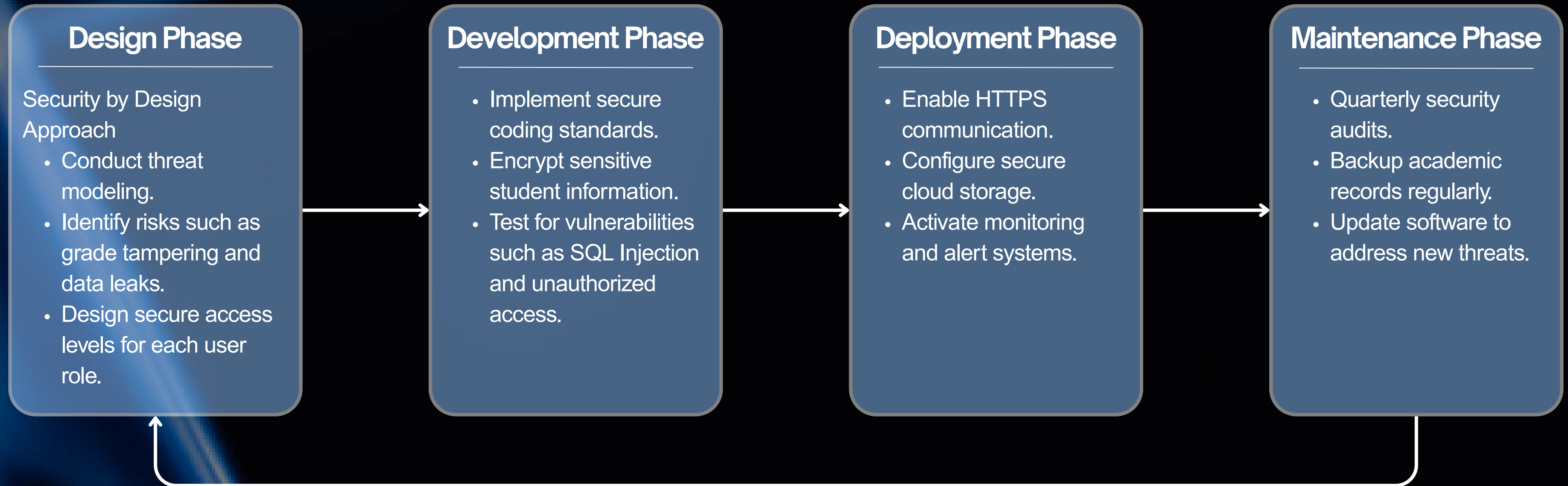
Security Objectives

- 1. Academic Integrity Protection:** Ensure that grades, attendance records, and academic reports cannot be altered without authorization.
- 2. Student Privacy Protection:** Safeguard personal information such as addresses, contact details, and academic history.
- 3. Accountability and Traceability:** Ensure every action in the system can be traced back to a specific user.

Security Requirements

- 1. Multi-Factor Authentication (MFA):** Teachers and administrators must verify their identity using a password plus a one-time code.
- 2. Audit Trail Logging:** Every change to grades, attendance, or student information is recorded with:
 - User ID
 - Date and time
 - Action performed
- 3. Role-Based Access Control:**
 - Students can only view their own records.
 - Parents can only view their child's records.
 - Teachers can only access assigned classes.
 - Administrators have broader access.

Security Assurance Across the Life Cycle



Build Security In



Why Security Must Be Built In Early

- Imagine security as the foundation of a school building.
- If the foundation is strong:
 - Student records remain protected.
 - Grade manipulation becomes difficult.
 - Compliance with data privacy laws is easier.
 - User trust increases.

Risks of Adding Security Later

Scenario:

- The system is already being used when administrators discover teachers can accidentally access records outside their department.
- Possible consequences:
 - Data privacy violations
 - Unauthorized grade modifications
 - Costly redesign of the system
 - School reputation damage

Trusted System Concepts

Reference Monitor

Grade Approval Gateway

- Before any grade modification is saved, the system checks:
 - Who is requesting the change?
 - Are they authorized?
 - Is the action allowed?
- **Only valid requests proceed.**

Security Kernel Components

1. **Identity Verification Module:** Verifies user credentials.
2. **Access Control Engine:** Determines permissions.
3. **Audit Logging Service:** Records all sensitive actions.
4. **Data Encryption Service:** Protects stored and transmitted information.

Trusted Computing Base (TCB)

The trusted components responsible for enforcing security:

- Authentication Server
- School Database
- Access Control Engine
- Audit Logging System
- Encryption Services
- Operating System Hosting the Application